

| Epic                                              | Story ID | Actor                       | User Stories                                                                                                                                                                                     | Acceptance Criteria (Given / When / Then)                                                                                                                                                                                                              |
|---------------------------------------------------|----------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1: System Administration & Health Data Management | US1.1    | WAH4PCE Admin               | As a WAH4PCE Admin, I want to administer the system using the IPaaS management console so that I can configure API endpoints, manage credentials, and monitor transaction logs and data mappings | Given the admin is authenticated, When they navigate the tabbed administration dashboard, Then they can view side-by-side JSON payloads (hashed or unhashed) and tabular comparisons of sent and received data.                                        |
|                                                   | US1.2    | WAH (WAH4S, WAH4C, WAH4H)   | As a WAH user, I want to manage health data so that accurate and updated patient records are securely provided to the interoperability network.                                                  | Given a WAH user needs to update or provide patient records, When they manage the health data in the system, Then the data is securely stored and prepared as the base for all external health data requests.                                          |
| 2: Health Data Request & Provisioning             | US2.1    | External (e.g., DOH iHomis) | As an External or WAH system, I want to bidirectionally request legacy HL7 health data so that I can securely retrieve specific patient records from the source organization.                    | Given an authenticated actor sends a health-data query, When the source organization's authorized user approves the request in their portal, Then the system extracts, transforms, and routes the HL7 response back to the requestor.                  |
|                                                   | US2.2    | WAH                         | As a WAH user, I want to request FHIR 4 health data so that I can retrieve modernized, fully standardized clinical information.                                                                  | Given a WAH facility needs to fetch a standardized patient record, When they execute a Request FHIR 4 Health Data command, Then the system extends the base request to ensure the returned data complies specifically with FHIR 4 standard formatting. |

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| 3: ADAPT Translation & Validation | US3.1 | System (Internal) | As the Internal System, I want to validate the schema and completeness of incoming JSON/XML payloads so that structural errors and missing routing fields are caught before semantic translation. | Given a data payload enters the core pipeline, When it undergoes internal validation, Then the system verifies required fields (source, target, format) and blocks processing if the payload is malformed.                         |
|                                   | US3.2 | ADAPT             | As the ADAPT system, I want to translate data between PH Core HL7 FHIR R4 and HL7 v2 formats using AI mapping rules so that cross-platform semantic interoperability is achieved.                 | Given a validated payload enters the ADAPT engine, When it applies AI mapping rules, Then it transforms the data, utilizing a deterministic algorithmic fallback if the AI transformation encounters an issue.                     |
|                                   | US3.3 | System (Internal) | As the Internal System, I want to validate patient privacy consent during translation so that no data is shared without adhering to the patient's active consent decisions.                       | Given the translation process is active, When the system reads the consent flag and registry, Then it allows the transaction if approved or quarantines the payload and returns an HTTP 422 error if consent is missing or denied. |